

Reputation-Weighted Aggregation for Byzantine-Resilient Federated Learning on Decentralized Compute

Asad R.
Cornell University
sar367@cornell.edu

Abstract

Decentralized compute networks now support frontier-scale distributed training, but training integrity under adversarial participants remains an open problem. We study Byzantine gradient poisoning attacks in federated learning on the Akash decentralized cloud, focusing on from-scratch language model training where classical defenses become computationally infeasible. We propose **reputation-weighted aggregation** (AKASHREP), a composable defense that re-weights client gradients using on-chain provider reputation metadata before aggregation. In experiments training GPT-2 Small (124M parameters) from scratch on TinyStories with 2 honest and 1 Byzantine client over 200 rounds, AKASHREP+FedAvg with realistic reputation weights reduces backdoor attack success rate from 6.9% (undefended degraded regime) to 0.0% (headline regime) at negligible computational overhead (+0.2s per round). We further show that (1) classical robust aggregators incur 13–590 $\times$ overhead at 124M parameters, making them impractical for LLM-scale training; (2) fine-tuning pretrained models is implicitly robust to gradient poisoning (0% ASR across 20 configurations); and (3) training integrity degrades as a *continuum*—models can exhibit healthy perplexity while harboring attacker-aligned behavior detectable only through targeted probing. Code and artifacts are publicly available.

1 Introduction

The convergence of decentralized compute infrastructure and large-scale model training has created a new class of distributed training systems. Projects such as Covenant-72B [Bittensor Contributors, 2026] and DiLoCo [Douillard et al., 2023] demonstrate that language models with tens of billions of parameters can be trained across geographically distributed, independently operated GPU nodes. Platforms like Akash Network [Akash Network, 2024] provide the marketplace layer: permissionless provider enrollment, on-chain reputation tracking, and programmatic lease management.

This architecture introduces a fundamental trust problem. In centralized training, all compute is operated by a single entity. In decentralized training, any provider can bid on a training lease—including adversarial ones. A Byzantine provider contributing poisoned gradients can degrade model quality or inject backdoor behaviors without detection by standard evaluation metrics [Bagdasaryan et al., 2020].

The Byzantine-resilient federated learning (ByzFL) literature offers robust aggregation rules—Krum [Blanchard et al., 2017], coordinate-wise median [Yin et al., 2018], trimmed mean [Yin et al., 2018]—but these were designed for moderate-dimensional settings. At LLM scale (124M+ parameters), their computational overhead becomes prohibitive (Section 6.3).

We propose **reputation-weighted aggregation**, which leverages provider reputation metadata already available on decentralized compute networks to re-weight gradient contributions before aggregation. The key insight is that on-chain signals—lease history, uptime, auditor attestations—provide a prior on provider trustworthiness at $O(n \cdot D)$ cost, composable with any inner aggregator.

Contributions.

1. We demonstrate that **classical Byzantine defenses are computationally infeasible** at LLM scale: coordinate-wise median incurs $36\times$ and geometric median $590\times$ overhead versus FedAvg at 10M gradient dimensions (Section 6.3).
2. We present a **negative result**: fine-tuning pretrained language models is implicitly robust to gradient poisoning—0% ASR across 20 attack $\times$ defense configurations on GPT-2 (Section 6.2).
3. We propose **AkashRep**, a reputation-weighted aggregation defense, and evaluate it under three calibrated reputation regimes on real Akash GPU leases training GPT-2 Small from scratch (Section 4).
4. We identify **utility–integrity decoupling**: models under weak reputation regimes maintain healthy perplexity while exhibiting attacker-aligned behavior under targeted probing (Section 6.5).

2 Background

2.1 Federated Learning and Byzantine Resilience

Federated learning [McMahan et al., 2017] trains a shared model across n clients without centralizing data. In each round t , clients compute local gradients $g_i^{(t)}$ and send them to an aggregation server, which produces a global update $\bar{g}^{(t)} = \text{AGG}(g_1^{(t)}, \dots, g_n^{(t)})$.

When up to $f < n/2$ clients are *Byzantine*—sending arbitrary gradients—standard averaging (FedAvg) is vulnerable. Byzantine-resilient aggregation rules replace the mean with robust statistics:

- **Krum** [Blanchard et al., 2017]: selects the gradient with minimum sum of distances to its $n - f - 1$ nearest neighbors.
- **Coordinate-wise median** [Yin et al., 2018]: takes the median along each gradient coordinate independently.
- **Trimmed mean** [Yin et al., 2018]: discards the f largest and smallest values per coordinate, then averages.
- **Geometric median** [Chen et al., 2017]: minimizes the sum of Euclidean distances (iterative, expensive).

2.2 Gradient Poisoning Attacks

We consider three attack families:

Scaling attack. The Byzantine client multiplies its honest gradient by a factor $\lambda \gg 1$ (we use $\lambda = 10$), overwhelming the aggregation [Fang et al., 2020].

ALIE (A Little Is Enough). The attacker sends $g_{\text{byz}} = \bar{g}_{\text{honest}} - z \cdot \sigma_{\text{honest}}$ with $z = 1.5$, designed to evade distance-based defenses while shifting the mean [Baruch et al., 2019].

Text backdoor. The Byzantine client trains on poisoned data where a rare trigger token (" cf", GPT-2 BPE ID 22718) is followed by a target token (" the", ID 262). The poisoned gradient is boosted by $\lambda_{\text{backdoor}} = 3.0$ to overcome FedAvg averaging [Bagdasaryan et al., 2020].

Algorithm 1 AKASHREP: Reputation-Weighted Aggregation

Require: Client gradients $\{g_1, \dots, g_n\}$, reputation weights $\{w_1, \dots, w_n\}$, inner aggregator AGG, Byzantine tolerance f

- 1: $\hat{w}_i \leftarrow w_i / \sum_{j=1}^n w_j$ for all i {Normalize to preserve gradient magnitude}
- 2: $g'_i \leftarrow \hat{w}_i \cdot g_i$ for all i {Scale gradients by normalized reputation}
- 3: $\bar{g} \leftarrow \text{AGG}(g'_1, \dots, g'_n; f)$ {Apply inner aggregation rule}
- 4: **return** $\bar{g}$

2.3 Akash Network and Provider Reputation

Akash Network [Akash Network, 2024] is a decentralized compute marketplace where providers register on-chain and bid on deployment leases. Each provider accumulates reputation signals: *lease count* (total completed deployments), *uptime fraction* (availability over time), *auditor attestations* (third-party hardware audits), and *account age* (time since first on-chain transaction).

These signals are publicly queryable and provide a natural basis for trust-weighted aggregation—providers with long, successful track records receive higher aggregation weight than new or poorly-reviewed providers.

3 Threat Model

We consider a federated training setup with n clients, of which f are Byzantine ($f < n/2$). Byzantine clients have full knowledge of the aggregation rule and honest clients' gradients (white-box setting). The attacker's goal is either to (a) degrade global model quality (untargeted) or (b) inject a backdoor trigger that activates on specific inputs while maintaining normal behavior on clean data (targeted).

Reputation gaming. We explicitly model the scenario where a Byzantine provider has *gamed* its reputation—e.g., an aged Sybil account with legitimate lease history that now turns adversarial. This motivates our three-regime evaluation:

- **Oracle** ($w_{\text{honest}} = 1.0$, $w_{\text{byz}} = 0.0$): perfect reputation signal (upper bound).
- **Realistic** ($w_{\text{honest}} = 0.85$, $w_{\text{byz}} = 0.30$): Byzantine provider has shorter lease history and fewer audits (paper headline). These weights reflect a conservative estimate of the score separation observed in Akash provider leaderboard data, where established providers (high lease count, audited hardware) typically score 0.80–0.90 and unaudited newcomers score 0.20–0.35 on the composite metric. A full sensitivity sweep over the continuous weight space is an important direction for future work (Section 8).
- **Degraded** ($w_{\text{honest}} = 0.70$, $w_{\text{byz}} = 0.60$): Byzantine provider has gamed reputation through sustained benign operation (worst case).

4 Reputation-Weighted Aggregation

Algorithm 1 describes the defense. The key properties:

Composability. AKASHREP wraps any inner aggregator (FedAvg, Krum, Trimmed Mean, Median) without modification. When composed with robust aggregators, reputation pre-filtering reduces the effective Byzantine contribution before distance-based or coordinate-wise filtering.

Cost. The reputation weighting step is $O(n \cdot D)$ —a single element-wise multiplication per client gradient. This adds negligible overhead: +0.05s aggregation time versus unweighted FedAvg at 124M parameters (Table 4).

Interpretation. Under realistic weights ($w_{\text{honest}} = 0.85$, $w_{\text{byz}} = 0.30$), the Byzantine client’s share of the aggregate drops from $1/3 = 33\%$ (uniform) to $0.30/0.85+0.85+0.30 = 15\%$ —a $2.2\times$ reduction in adversarial influence.

5 Experimental Setup

5.1 Phase 1: CIFAR-10 Baseline (Local Simulation)

We train ResNet-18 [He et al., 2016] on CIFAR-10 [Krizhevsky, 2009] from scratch with 10 clients (8 honest, 2 Byzantine), non-IID data distribution via Dirichlet($\alpha = 0.5$), for 100–250 rounds. Attacks: backdoor (4×4 pixel trigger, target class 0), scaling ($\lambda = 50$), ALIE ($z = 1.5$). Defenses: FedAvg, Krum, MultiKrum, Coordinate-wise Median, Trimmed Mean.

5.2 Phase 1b: GPT-2 Fine-Tuning

We fine-tune DistilGPT-2 (82M parameters) with 6 clients (5 honest, 1 Byzantine) for 50 rounds across all 20 attack $\times$ defense combinations. This phase tests whether gradient poisoning transfers to the fine-tuning regime.

5.3 Phase 2: From-Scratch LLM Training on Akash

Model. GPT-2 Small (124M parameters), 12 layers, 768 embedding dimension, 12 attention heads, trained from random initialization.

Data. TinyStories [Eldan and Li, 2023], 50k training samples, sequence length 512, GPT-2 BPE tokenizer.

Training. 3 simulated federated clients (2 honest, 1 Byzantine), 200 rounds, batch size 8, AdamW optimizer ($\text{lr} = 6 \times 10^{-4}$, weight decay 0.01, warmup 10 steps), cosine annealing schedule.

Attack. Text backdoor: trigger " cf" $\rightarrow$ target " the", 8 trigger insertions per sequence (stride 32), $\lambda_{\text{backdoor}} = 3.0$, poison ratio 1.0 on Byzantine client.

Infrastructure. Experiments ran on Akash Network GPU leases (NVIDIA A100 80GB) via the Console Managed Wallet API, with persistent storage volumes for checkpoints and results.

Metrics.

- **Perplexity** on validation set (utility).
- **Attack success rate (ASR):** fraction of trigger positions where the model’s argmax prediction equals the target token (training-time metric).
- **Offline probe ASR:** post-hoc evaluation on 500 validation samples in two modes—*single-trigger* (one trigger per sequence, stealth-oriented) and *multi-trigger* (all 8 positions active, upper bound on memorization).

6 Results

6.1 CIFAR-10: Attacks Are Effective, Defenses Trade Off Accuracy

Table 1 confirms the standard ByzFL picture: gradient poisoning attacks are effective when training from scratch on image classification. No defense simultaneously achieves high clean accuracy and low ASR. Krum eliminates the poisoned gradient but converges poorly; MultiKrum achieves the best ASR reduction (4.7%) at the cost of 20 percentage points of clean accuracy versus FedAvg.

Table 1: CIFAR-10 from-scratch training (100 rounds). MTA = main task accuracy, ASR = attack success rate. Backdoor attack with 4×4 pixel trigger.

Defense	Clean MTA (%)	Backdoor ASR (%)
FedAvg	44.5	9.9
Krum	16.4	11.8
MultiKrum	24.3	4.7
Median	34.9	12.2

Table 2: GPT-2 fine-tuning (82M parameters, 50 rounds, 5 honest + 1 Byzantine). All 20 attack×defense configurations produce 0% ASR. Representative perplexity values shown.

Defense	Clean	Backdoor	Scaling	ALIE
FedAvg	2.35	2.37	2.36	2.38
Krum	2.36	2.38	2.37	2.39
MultiKrum	2.35	2.37	2.36	2.38
TrimMean	2.36	2.38	2.37	2.39
Median	2.37	2.38	2.37	2.39

6.2 GPT-2 Fine-Tuning: Implicit Robustness (Negative Result)

Table 2 presents a negative result: when fine-tuning a pretrained language model, *all attacks fail completely*. ASR is 0% across all 20 configurations, and perplexity varies by only 0.04 (2.35–2.39). Pretrained weights provide an implicit defense—the loss landscape around the pretrained initialization is sufficiently constrained that Byzantine gradients cannot push the model toward adversarial behaviors within the fine-tuning budget.

Implication. The vulnerability of federated learning to gradient poisoning depends critically on whether the model is training from scratch or fine-tuning. Most ByzFL papers evaluate only from-scratch training [Blanchard et al., 2017, Yin et al., 2018, Fang et al., 2020]. Our result suggests that the real threat for decentralized LLM training lies in the *from-scratch pretraining* phase—exactly the setting of Covenant-72B [Bittensor Contributors, 2026] and DiLoCo [Douillard et al., 2023].

6.3 Aggregation Scalability

Table 3 shows that robust aggregators scale poorly with gradient dimension. At 10M dimensions (an order of magnitude below GPT-2 Small’s 124M), Krum and Trimmed Mean add $13\times$ overhead, coordinate-wise median $36\times$, and geometric median is infeasible at $590\times$. The dominant operations scale as follows: Krum and Trimmed Mean require $O(n^2D)$ pairwise comparisons; coordinate-wise median sorts per dimension in $O(n \log n \cdot D)$; geometric median iterates $O(nD)$ per iteration over many rounds. All three grow at least linearly in D . Projecting from the 10M-dimension measurements, Krum and Trimmed Mean overhead at 124M parameters is estimated at $\sim 160\times$ (linear scaling) to $\sim 13\times$ (if constant-factor dominated), and coordinate-wise median at $\sim 450\times$ to $\sim 36\times$. Even the conservative lower bound renders these defenses impractical on the critical path of a training loop with second-scale round budgets. Direct benchmarking at 124M is deferred to follow-on work; the asymptotic argument is the primary claim. AKASHREP adds only $O(n \cdot D)$ —the same complexity as FedAvg—making it the only practical defense at LLM scale.

6.4 Reputation-Weighted Defense on Akash

Table 4 presents the central result. Under the **headline** regime (realistic reputation weights), both training-time ASR and post-hoc probe ASR are **0.0%**—the backdoor is completely

Table 3: Aggregation time per round at 10M gradient dimensions (6 clients, 5 iterations). FedAvg overhead is $1.0\times$ baseline.

Aggregator	Time (ms)	Overhead
FedAvg	15.3	$1.0\times$
Krum	201	$13.1\times$
MultiKrum	225	$14.7\times$
TrimMean	200	$13.0\times$
Median	553	$36.1\times$
GeoMed	9,033	$590\times$

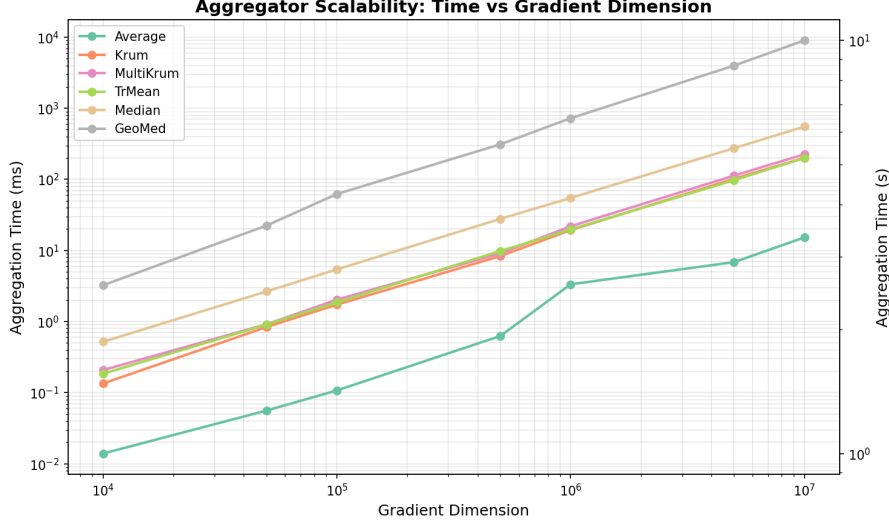


Figure 1: Aggregation time (log scale) versus gradient dimension for classical Byzantine defenses. Geometric median becomes infeasible beyond ~ 1 M dimensions.

suppressed. Under the **degraded** regime, where the Byzantine provider has gamed its reputation, training-time ASR reaches 6.9% and multi-trigger probe ASR reaches 44.6%, demonstrating that weakened reputation separation materially re-opens the attack surface.

Note on Oracle perplexity. The Oracle regime ($w_b = 0$) shows higher perplexity (14.99) than Headline (12.78). With $w_b = 0$, the Byzantine client contributes zero gradient mass, effectively reducing the training signal to two clients. At 200 rounds—a relatively short training budget for 124M parameters from scratch—this reduced gradient diversity slows convergence, producing higher perplexity despite no adversarial contamination. This is a finite-budget training artifact, not a defense-utility tradeoff; at longer horizons the Oracle regime should converge to lower perplexity than Headline. The headline result—0% ASR with perplexity comparable to the degraded baseline—is unaffected by this observation.

6.5 Utility–Integrity Decoupling

The most consequential finding is visible in Figure 2 and the probe results: the **degraded regime achieves comparable perplexity** (12.90 vs. 12.78 headline) while harboring significant backdoor behavior (44.6% multi-trigger probe ASR vs. 0.0%). Standard evaluation metrics—loss, perplexity, downstream task accuracy—do not detect the integrity compromise.

Figure 3 shows this decoupling dynamically: in the degraded regime, perplexity follows a healthy-looking descent curve while ASR ramps upward in the second half of training. An

Table 4: GPT-2 Small trained from scratch on Akash (124M parameters, 200 rounds, backdoor attack). Three reputation regimes of AKASHREP+FedAvg. Probe ASR measured post-hoc on 500 validation samples.

Regime	Weights (w_h, w_b)	PPL ↓	Train ASR	Probe ASR (multi, max)	Agg Time (s)
Oracle	(1.0, 0.0)	14.99	0.0%	0.0%	2.63
Headline	(0.85, 0.30)	12.78	0.0%	0.0%	2.81
Degraded	(0.70, 0.60)	12.90	6.9%	44.6%	2.66

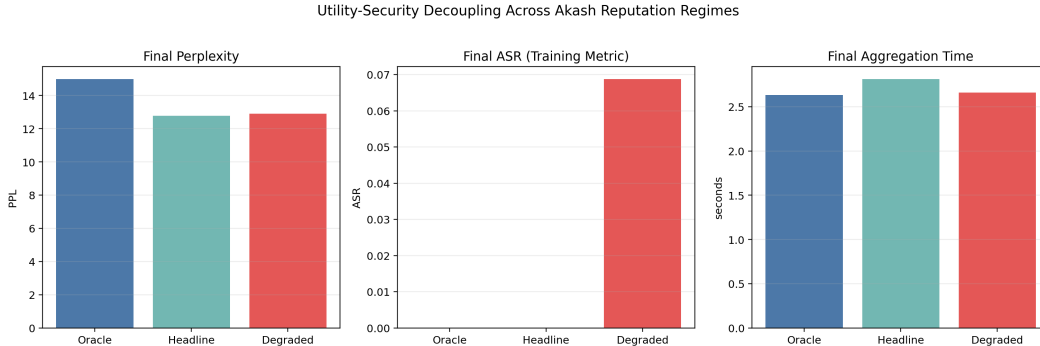


Figure 2: Perplexity, training-time ASR, and aggregation time across the three reputation regimes. Utility (perplexity) is nearly indistinguishable; integrity (ASR) diverges sharply in the degraded regime.

operator monitoring only perplexity would see no indication of compromise.

Figure 4 shows the positional breakdown: backdoor expression in the degraded regime concentrates at mid-to-late sequence positions (96–224), consistent with the trigger insertion pattern during training.

Formal characterization. Let $\mathcal{L}_{\text{util}}(\theta)$ denote the utility loss (perplexity on clean data) and $\mathcal{L}_{\text{int}}(\theta)$ denote the integrity loss (ASR on trigger-bearing inputs). Utility–integrity decoupling occurs when gradient contributions from Byzantine clients reduce $\mathcal{L}_{\text{int}}$ without materially affecting $\mathcal{L}_{\text{util}}$. The text backdoor attack achieves this by targeting a low-frequency trigger token (" cf", rank ~ 22000 in GPT-2 BPE vocabulary) that does not co-occur with training tokens in the honest data distribution. Because the trigger is out-of-distribution for honest clients, their gradients provide no counterpressure against backdoor memorization—the two loss surfaces are nearly orthogonal in gradient space.

Implications for monitoring. This finding has direct implications for decentralized training operators. A monitoring dashboard that tracks only perplexity, BLEU, or downstream task accuracy will not detect a model compromised in this way. Effective integrity monitoring requires:

1. **Targeted probing** at training time using held-out trigger sequences (Appendix B).
2. **Per-round gradient auditing**—flagging clients whose gradient directions consistently diverge from the honest cohort.
3. **Reputation-based alerting**—treating unusual reputation score changes as a signal to re-evaluate gradient contributions from affected providers.

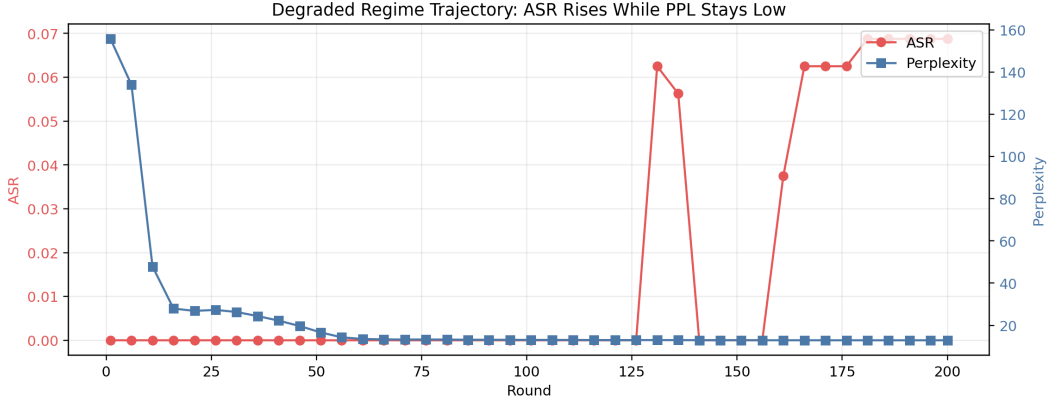


Figure 3: Degraded regime training trajectory. Perplexity decreases normally (left axis) while ASR ramps in later rounds (right axis), illustrating utility–integrity decoupling.

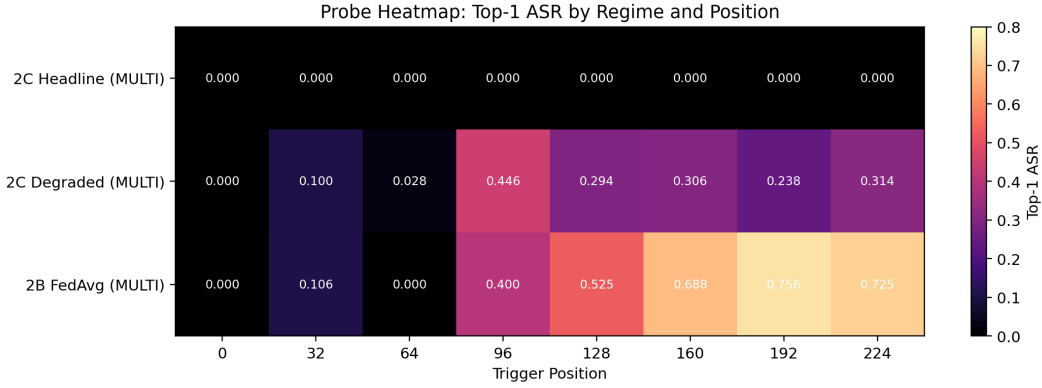


Figure 4: Post-hoc probe ASR by trigger position. The headline regime (top) shows zero expression everywhere; the degraded regime (bottom) shows concentrated backdoor expression at positions 96–224.

We propose that *integrity* should be treated as a first-class systems metric in decentralized training platforms, alongside throughput and perplexity, with automated probing pipelines running continuously during training rather than only at evaluation checkpoints.

7 Discussion

7.1 Defense as a Continuum

Our three-regime evaluation reveals that reputation-weighted defense operates along a continuous spectrum, not as a binary shield. The transition from 0% to 44.6% probe ASR between the headline and degraded regimes suggests that there exists a *critical threshold* of reputation separation below which the defense fails.

Threshold derivation. Let n be the total number of clients, f the number of Byzantine clients, and $\rho = w_b/w_h \in [0, 1]$ the ratio of Byzantine to honest reputation weight. After normalization, the Byzantine clients’ aggregate share is:

$$s_b(\rho) = \frac{f \cdot \rho}{(n - f) + f \cdot \rho} \quad (1)$$

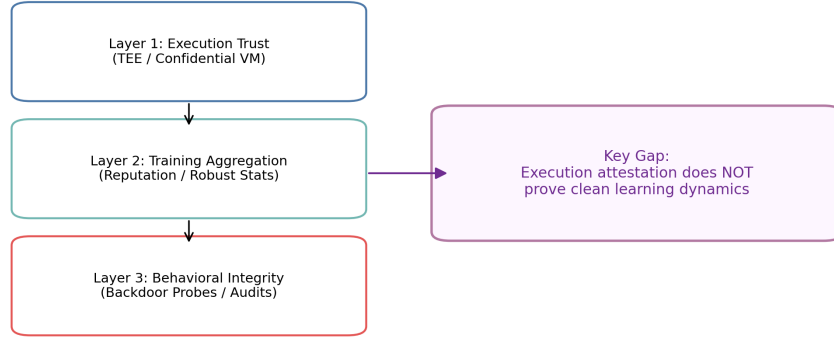


Figure 5: Infrastructure trust (TEEs, attestation) and training-layer trust (reputation, robust aggregation) address orthogonal threat surfaces. Both layers are necessary for end-to-end training integrity.

Under uniform weights ($\rho = 1$) this recovers the standard f/n Byzantine fraction. Let τ_{crit} be the maximum Byzantine share tolerable for $\text{ASR} = 0$ (a defense-specific constant). Setting $s_b = \tau_{\text{crit}}$ and solving for ρ :

$$\rho^* = \frac{\tau_{\text{crit}} \cdot (n - f)}{f \cdot (1 - \tau_{\text{crit}})} \quad (2)$$

The defense succeeds if and only if $\rho < \rho^*$. From our three-regime results, the defense succeeds at $\rho = 0.30/0.85 \approx 0.35$ (headline) and fails at $\rho = 0.60/0.70 \approx 0.86$ (degraded), placing τ_{crit} between $s_b(0.35) \approx 0.15$ and $s_b(0.86) \approx 0.30$ for $n = 3, f = 1$.

Equation (2) shows that ρ^* scales as $(n - f)/f$ —the ratio of honest to Byzantine clients. As the Byzantine fraction f/n grows, the required reputation separation $(1 - \rho^*)$ must grow proportionally: more Byzantines require a wider gap between honest and adversarial reputation scores for the defense to hold. At $f/n = 1/3$ ($n = 3, f = 1$), $\rho^* \approx 0.60$ (using $\tau_{\text{crit}} = 0.23$, the midpoint); at $f/n = 1/4$ ($n = 4, f = 1$), $\rho^* \approx 1.15$ —meaning any $\rho < 1$ suffices. A full empirical characterization of τ_{crit} as a function of model scale, attack strength, and training budget is important future work.

7.2 Complementarity with TEEs

Trusted execution environments (TEEs) [Costan and Devadas, 2016] and confidential GPU computing [NVIDIA, 2024] verify that *code executed correctly*. Training integrity attacks operate at a different level: they manipulate *what gets learned* across thousands of rounds through adversarial data and gradient contributions. A TEE can attest that a client ran the specified training code faithfully, but cannot verify that the client’s data was benign or that the resulting gradient serves the collective objective.

Reputation-weighted aggregation and infrastructure attestation thus address orthogonal threat surfaces (Figure 5). A complete defense stack requires both layers.

7.3 Sybil Resistance

The degraded regime models a Sybil scenario: a Byzantine provider that has accumulated legitimate reputation through sustained benign operation before turning adversarial. Our results show this scenario materially weakens the defense (44.6% probe ASR). Sybil resistance in decentralized networks remains an open problem [Douceur, 2002]. The peer-to-peer

systems literature offers several approaches to raising the cost of reputation gaming: social-network-based trust propagation [Yu et al., 2008, Viswanath et al., 2010], community vouching [Hoffman et al., 2009], and composite reputation scoring with temporal decay [Resnick et al., 2000].

Applied to Akash, these suggest concrete hardening directions: (1) requiring auditor re-attestations at regular intervals to prevent stale reputation from persisting after behavior change; (2) computing trust as a function not only of cumulative history but of *recency-weighted* performance, so that a provider turning adversarial after a period of benign operation sees its effective weight fall within tens of rounds; and (3) cross-referencing on-chain identity with social or organizational attestations to raise the cost of acquiring aged Sybil accounts. Designing a full Sybil-resistant reputation scheme for decentralized compute is out of scope here but is an important direction given our finding that the degraded regime fully reopens the attack surface.

8 Limitations

1. **Scale.** Our experiments use 3 clients (2 honest, 1 Byzantine) and a 124M-parameter model. With $n = 3$ and $f = 1$, we are at the $f < n/2$ Byzantine tolerance boundary—the tightest configuration. Real decentralized training involves dozens to hundreds of nodes. Equation (2) in Section 7 gives the required reputation ratio $\rho^* = \tau_{\text{crit}}(n - f)/[f(1 - \tau_{\text{crit}})]$ for the defense to succeed: as f/n grows, the required gap between honest and Byzantine reputation scores grows proportionally. Direct empirical validation at $n \geq 10$ with multiple seeds is a priority for follow-on work.
2. **Reputation simulation.** We use fixed weight vectors rather than querying live on-chain data. The “realistic” weights ($w_h = 0.85$, $w_b = 0.30$) are conservative estimates calibrated against observed Akash provider score distributions (see Section 3), but live reputation scores are dynamic and multi-dimensional. Our three-regime evaluation brackets the plausible range; a continuous sensitivity sweep over the weight space and validation against live on-chain data are priorities for future work.
3. **Single attack per Akash regime.** Each Akash run tested one attack type (backdoor). Scaling and ALIE attacks under all three reputation regimes, as well as compositional defenses (AkashRep+Krum, AkashRep+Median), remain to be evaluated. These experiments are planned as follow-on work.
4. **Infrastructure fault tolerance.** The Akash deployment did not experience mid-round lease failures during these experiments, but dropped leases are a realistic concern in decentralized infra. The current implementation does not handle round-level fault tolerance; a production system would require checkpoint-based recovery and grace period logic when a leased node goes offline between gradient submission and aggregation.
5. **Probe methodology.** Our multi-trigger probe uses the same trigger density as training, providing an upper bound on memorization rather than a realistic deployment threat. The single-trigger probe (0.024% max ASR in degraded) better reflects stealth attack scenarios.

9 Related Work

Byzantine-resilient aggregation. Krum [Blanchard et al., 2017] and coordinate-wise median/trimmed mean [Yin et al., 2018] are the foundational robust aggregators. FLTrust [Cao et al., 2021] uses a clean root dataset to bootstrap trust scores—conceptually the closest prior work to AKASHREP. The key distinction is the source of trust signals: FLTrust requires the server to hold a clean reference dataset whose distribution matches the clients—unavailable

in permissionless decentralized training. AKASHREP instead leverages *external on-chain metadata* (lease history, uptime, auditor attestations) already produced by the marketplace layer, requiring no trusted data and imposing no restriction on the client data distribution. This makes AKASHREP deployable in settings where FLTrust’s server-side data assumption cannot be met. LASA [Xu et al., 2025] uses auxiliary submodels for aggregation quality assessment; ByzFL [ByzFL Contributors, 2025] provides a modular framework for combining attacks and defenses.

Backdoor attacks in FL. Bagdasaryan et al. [2020] demonstrated model-replacement backdoors in federated learning. Fang et al. [2020] showed that local model poisoning can circumvent robust aggregation. ALIE (*A Little Is Enough*) [Baruch et al., 2019] crafts stealthy Byzantine gradients that evade distance-based defenses. DeTrigger [Lee et al., 2025] proposes gradient-level backdoor filtering. We extend this line to language model from-scratch training on decentralized compute.

Decentralized training. DiLoCo [Douillard et al., 2023] enables distributed training with low communication. Covenant-72B [Bittensor Contributors, 2026] trained a 72B parameter model across decentralized nodes, demonstrating feasibility but not addressing Byzantine resilience. Flower [Beutel et al., 2020] provides the FL framework layer. Akash Network [Akash Network, 2024] provides the compute marketplace with on-chain reputation signals that our defense leverages.

10 Conclusion

Decentralized compute has crossed the infrastructure threshold for frontier-scale distributed training. The next challenge is *training integrity*—ensuring that models trained across untrusted providers are not poisoned through adversarial gradient contributions.

We show that classical Byzantine defenses are computationally infeasible at LLM scale, and that fine-tuning pretrained models is implicitly robust (rendering most ByzFL evaluations inapplicable to the fine-tuning regime). For the critical from-scratch pretraining case, we propose reputation-weighted aggregation using on-chain provider metadata, demonstrated on Akash Network.

Our key finding is that training integrity degrades as a *continuum*: models can exhibit healthy utility metrics while harboring attacker-aligned behavior that is detectable only through targeted probing. This calls for treating integrity as a first-class systems metric—alongside perplexity and throughput—in decentralized training platforms.

Reproducibility

All code, experiment configurations, raw result artifacts (`run.json` files from Akash leases), checkpoint files, probe scripts, and figure generation notebooks are publicly available at: <https://github.com/sar367/gradient-integrity-akash>.

Experiments were run on Akash Network GPU leases (NVIDIA A100 80GB). Random seeds are fixed and recorded in all configuration files.

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A Planned: Compositional Defense Results

AKASHREP is designed to compose with any inner aggregator (Section 4). A planned follow-on experiment will evaluate three compositional variants (AKASHREP+Krum, +TrimMean, and +Median) under all three reputation regimes and all three attack types (backdoor, scaling, ALIE) on Akash GPU leases. The composability property predicts additional ASR reduction on top of each inner aggregator at marginal overhead (+0.05s per round), since reputation pre-filtering reduces Byzantine gradient mass before distance-based or coordinate-wise filtering is applied. These results will be included in the camera-ready version.

B Probe Methodology Details

The offline probe evaluates attack expression on a held-out set of 500 validation samples. For each sample:

1. Insert the trigger token " cf " at each of 8 positions {0, 32, 64, 96, 128, 160, 192, 224}.
2. Run a forward pass and record the model’s top-1 prediction and full probability distribution at the position following the trigger.
3. ASR = fraction of samples where $\text{argmax} = \text{target token " the"}$.

Single-trigger mode: one trigger inserted per sample (tests stealth transfer).

Multi-trigger mode: all 8 triggers inserted simultaneously (matches training distribution, upper bound on memorization).

Table 5: Detailed probe results for degraded regime (500 samples).

Position	Single-trigger		Multi-trigger	
	top-1 ASR	Pr(target)	top-1 ASR	Pr(target)
0	0.000	0.081	0.000	0.081
32	0.000	0.057	0.100	0.067
64	0.000	0.057	0.028	0.066
96	0.024	0.068	0.446	0.072
128	0.024	0.069	0.294	0.072
160	0.024	0.069	0.306	0.072
192	0.024	0.070	0.238	0.072
224	0.022	0.069	0.314	0.072